ColabFold v1.6.0: AlphaFold2 using MMseqs2

Easy to use protein structure and complex prediction using AlphaFold2 and Alphafold2-multimer. Sequence alignments/templates are generated through MMseqs2 and HHsearch. For more details, see bottom of the notebook, checkout the ColabFold GitHub and Nature Protocols.

Old versions: v1.4, v1.5.1, v1.5.2, v1.5.3-patch

Mirdita M, Schütze K, Moriwaki Y, Heo L, Ovchinnikov S, Steinegger M. ColabFold: Making protein folding accessible to all. *Nature Methods*, 2022

Input protein sequence(s), then hit `Runtime` -> `Run all`

query_sequence

QDCQVSNFAVQQDFNRTRYQGTWYAVAKKDPVGLFLLDNIVANFKVEEDGTMTATAIGRVIILNNWE

- Use `:` to specify inter-protein chainbreaks for **modeling complexes** (supports homo- and hetro-oligomers). For example **PI...SK:PI...SK** for a homodimer

jobname

test

num_relax

0

- specify how many of the top ranked structures to relax using amber

template_mode

none

- `none` = no template information is used. `pdb100` = detect templates in pdb100 (see notes). `custom` - upload and search own templates (PDB or mmCIF format, see notes)

Show code

```
jobname test_3260b
sequence QDCQVSNFAVQQDFNRTRYQGTWYAVAKKDPVGLFLLDNIVANFKVEEDGTMTATAIGRVIILN
length 175
```

Install dependencies

Show code

```
installing colabfold...
CPU times: user 3.38 ms, sys: 55 µs, total: 3.44 ms
Wall time: 40 s
```

MSA options (custom MSA upload, single sequence, pairing mode)

msa_mode

mmseqs2_uniref_env

pair_mode

unpaired_paired

- "unpaired_paired" = pair sequences from same species + unpaired MSA, "unpaired" = seperate MSA for each chain, "paired" - only use paired sequences.

Show code

Advanced settings

model_type

auto

- if `auto` selected, will use `alphafold2_ptm` for monomer prediction and `alphafold2_multimer_v3` for complex prediction. Any of the mode_types can be used (regardless if input is monomer or complex).

num_recycles

3

- if `auto` selected, will use `num_recycles=20` if `model_type=alphafold2_multimer_v3`, else `num_recycles=3` .

recycle_early_stop_tolerance

auto

- if `auto` selected, will use `tol=0.5` if `model_type=alphafold2_multimer_v3` else `tol=0.0`.

relax_max_iterations

200

- max amber relax iterations, `0` = unlimited (AlphaFold2 default, can take very long)

pairing_strategy

greedy

- `greedy` = pair any taxonomically matching subsets, `complete` = all sequences have to match in one line.

☐ calc_extra_ptm

- return pairwise chain iptm/actifptm

Sample settings

- enable dropouts and increase number of seeds to sample predictions from uncertainty of the model.
- decrease `max_msa` to increase uncertainity

max_msa

auto

num_seeds

1

☐ use_dropout

Save settings

- [ ] save_all
- [ ] save_recycles
- [ ] save_to_google_drive

- if the save_to_google_drive option was selected, the result zip will be uploaded to your Google Drive

dpi

200

- set dpi for image resolution

Don't forget to hit `Runtime` -> `Run all` after updating the form.

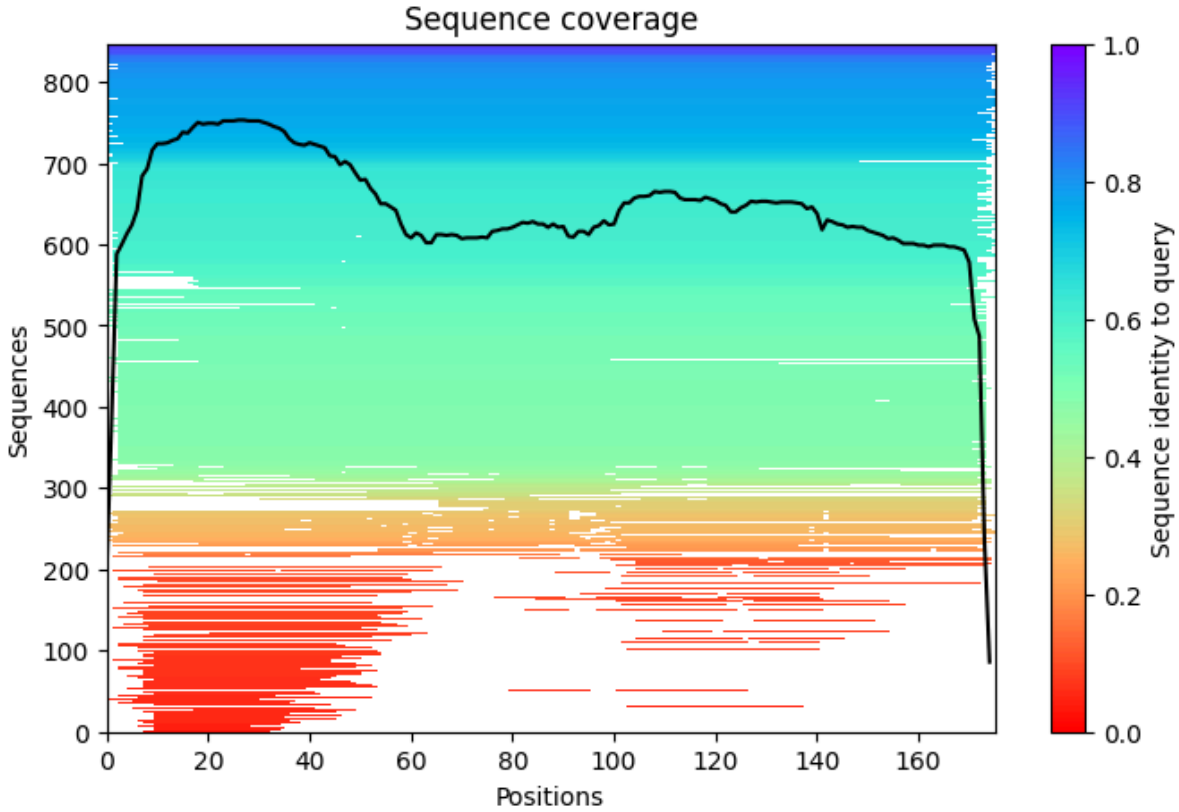
Show code

Run Prediction

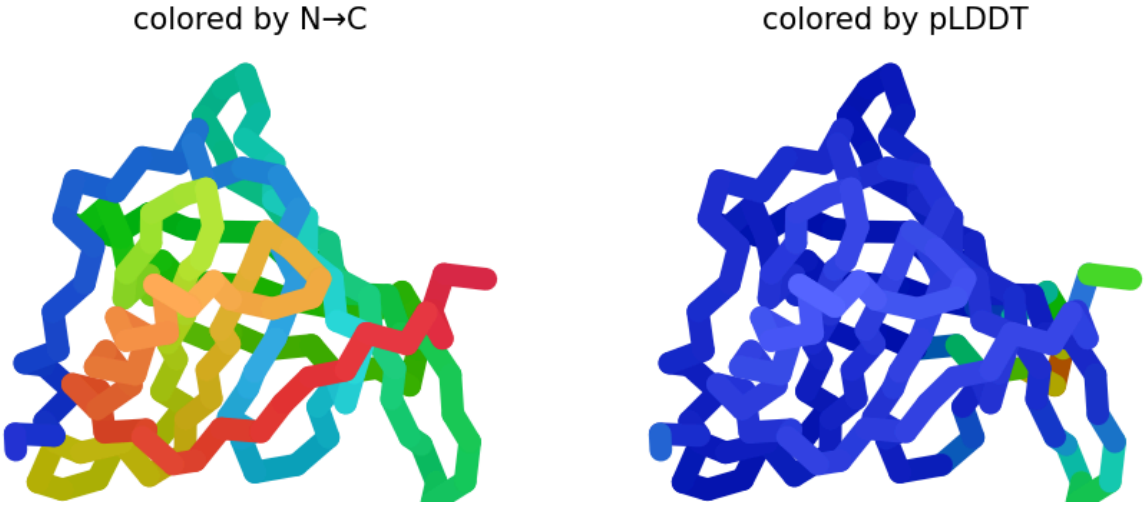
display_images

Show code

```
Downloading alphafold2_ptm weights to .: 100%|██████████| 3.47G/3.47G [(
2026-03-11 18:13:38,104 Running on GPU
2026-03-11 18:13:38,424 Found 5 citations for tools or databases
2026-03-11 18:13:38,425 Query 1/1: test_3260b (length 175)
PENDING:   0%|          | 0/150 [elapsed: 00:00 remaining: ?]2026-03-11
RUNNING:   5%|▌         | 7/150 [elapsed: 00:08 remaining: 02:45]2026-03
RUNNING:   9%|▉         | 14/150 [elapsed: 00:15 remaining: 02:31]2026-0
RUNNING:  15%|▉         | 23/150 [elapsed: 00:25 remaining: 02:17]2026-0
RUNNING:  19%|▉         | 28/150 [elapsed: 00:30 remaining: 02:13]2026-0
RUNNING:  24%|▉         | 36/150 [elapsed: 00:39 remaining: 02:03]2026-0
COMPLETE: 100%|██████████| 150/150 [elapsed: 00:49 remaining: 00:00]
```

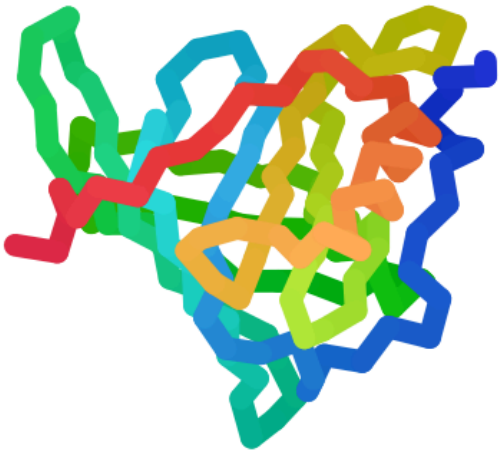



```
2026-03-11 18:14:29,183 Setting max_seq=512, max_extra_seq=335
2026-03-11 18:15:16,640 alphafold2_ptm_model_1_seed_000 recycle=0 pLDDT=
2026-03-11 18:15:47,877 alphafold2_ptm_model_1_seed_000 recycle=1 pLDDT=
2026-03-11 18:15:58,687 alphafold2_ptm_model_1_seed_000 recycle=2 pLDDT=
2026-03-11 18:16:09,675 alphafold2_ptm_model_1_seed_000 recycle=3 pLDDT=
2026-03-11 18:16:09,676 alphafold2_ptm_model_1_seed_000 took 87.7s (3 re
```

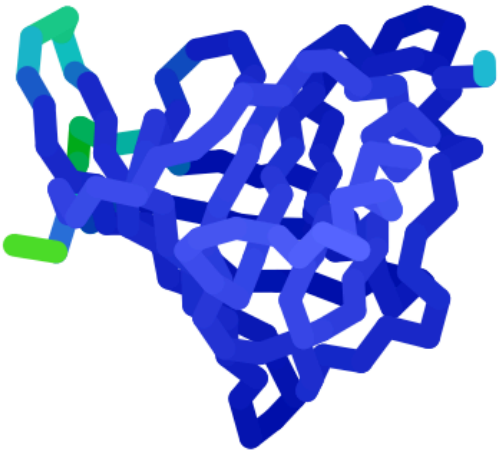


```
2026-03-11 18:16:20,965 alphafold2_ptm_model_2_seed_000 recycle=0 pLDDT=
2026-03-11 18:16:32,345 alphafold2_ptm_model_2_seed_000 recycle=1 pLDDT=
2026-03-11 18:16:43,938 alphafold2_ptm_model_2_seed_000 recycle=2 pLDDT=
2026-03-11 18:16:55,820 alphafold2_ptm_model_2_seed_000 recycle=3 pLDDT=
2026-03-11 18:16:55,821 alphafold2_ptm_model_2_seed_000 took 46.0s (3 re
```

colored by N→C

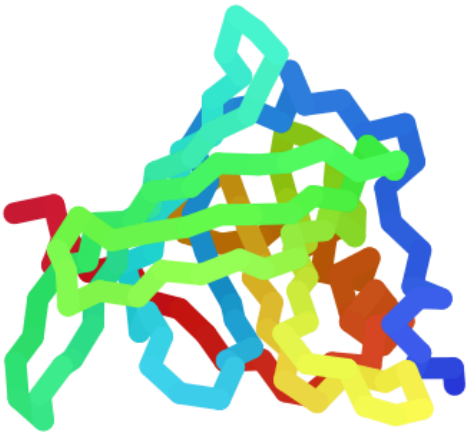


colored by pLDDT

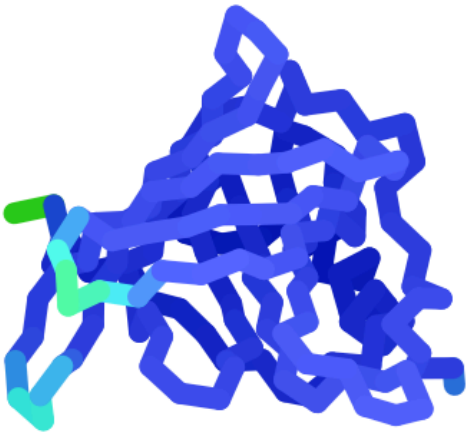


```
2026-03-11 18:17:07,761 alphafold2_ptm_model_3_seed_000 recycle=0 pLDDT=
2026-03-11 18:17:19,197 alphafold2_ptm_model_3_seed_000 recycle=1 pLDDT=
2026-03-11 18:17:30,473 alphafold2_ptm_model_3_seed_000 recycle=2 pLDDT=
2026-03-11 18:17:41,754 alphafold2_ptm_model_3_seed_000 recycle=3 pLDDT=
2026-03-11 18:17:41,755 alphafold2_ptm_model_3_seed_000 took 45.8s (3 re
```

colored by N→C



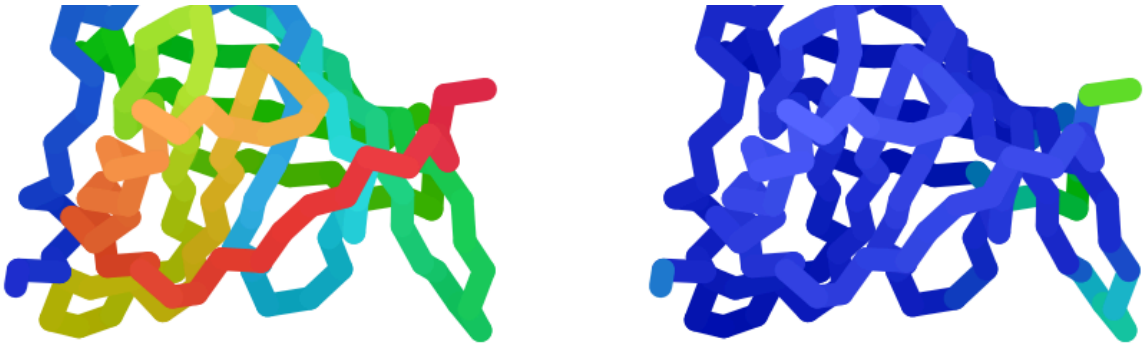
colored by pLDDT



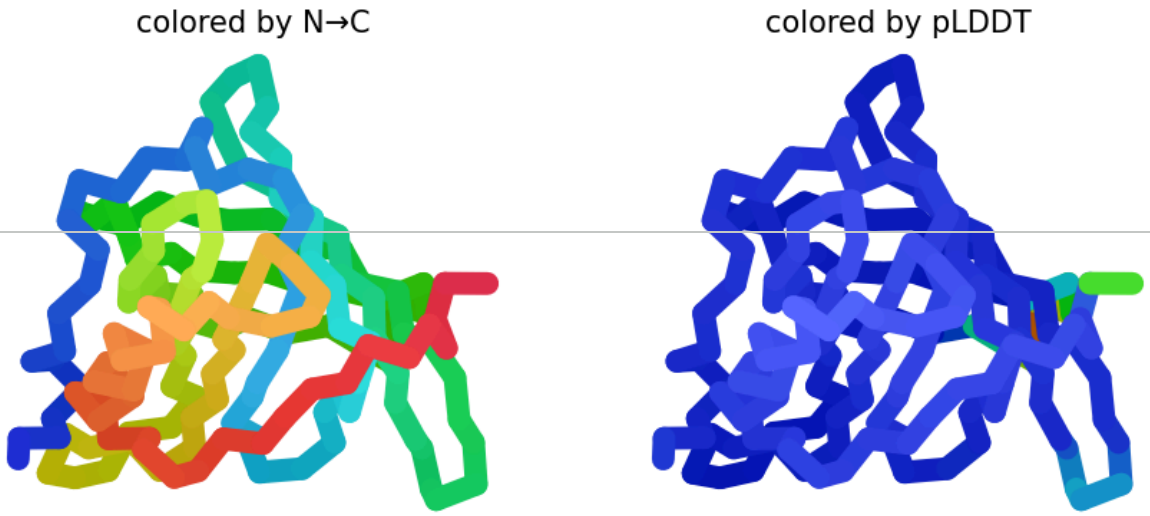
```
2026-03-11 18:17:53,248 alphafold2_ptm_model_4_seed_000 recycle=0 pLDDT=
2026-03-11 18:18:04,759 alphafold2_ptm_model_4_seed_000 recycle=1 pLDDT=
2026-03-11 18:18:16,326 alphafold2_ptm_model_4_seed_000 recycle=2 pLDDT=
2026-03-11 18:18:27,859 alphafold2_ptm_model_4_seed_000 recycle=3 pLDDT=
2026-03-11 18:18:27,860 alphafold2_ptm_model_4_seed_000 took 46.0s (3 re
```

colored by N→C

colored by pLDDT



```
2026-03-11 18:18:39,463 alphafold2_ptm_model_5_seed_000 recycle=0 pLDDT=
2026-03-11 18:18:50,907 alphafold2_ptm_model_5_seed_000 recycle=1 pLDDT=
2026-03-11 18:19:02,319 alphafold2_ptm_model_5_seed_000 recycle=2 pLDDT=
2026-03-11 18:19:13,741 alphafold2_ptm_model_5_seed_000 recycle=3 pLDDT=
2026-03-11 18:19:13,742 alphafold2_ptm_model_5_seed_000 took 45.8s (3 re
```




```
2026-03-11 18:19:13,894 reranking models by 'plddt' metric
2026-03-11 18:19:13,894 rank_001_alphafold2_ptm_model_3_seed_000 pLDDT=9
2026-03-11 18:19:13,895 rank_002_alphafold2_ptm_model_2_seed_000 pLDDT=9
2026-03-11 18:19:13,895 rank_003_alphafold2_ptm_model_4_seed_000 pLDDT=9
2026-03-11 18:19:13,895 rank_004_alphafold2_ptm_model_5_seed_000 pLDDT=9
2026-03-11 18:19:13,895 rank_005_alphafold2_ptm_model_1_seed_000 pLDDT=9
2026-03-11 18:19:15,077 Done
0
```

Display 3D structure

rank_num

1

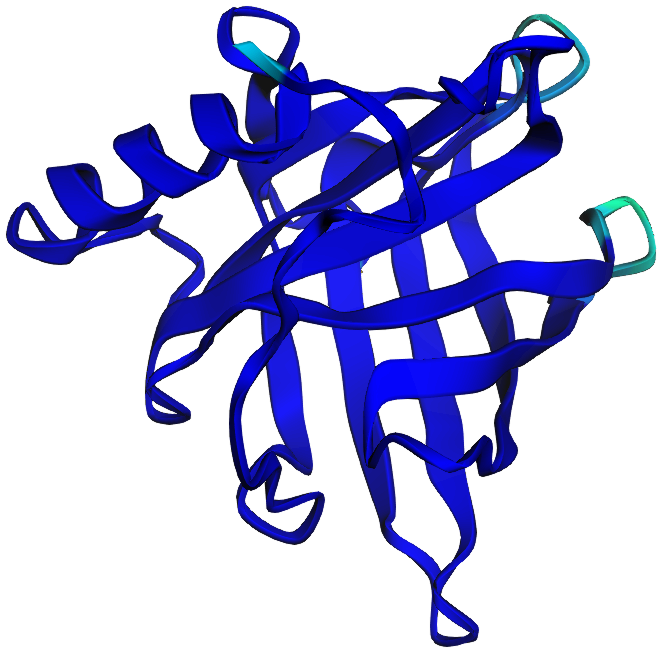
color

lDDT

show_sidechains

show_mainchains

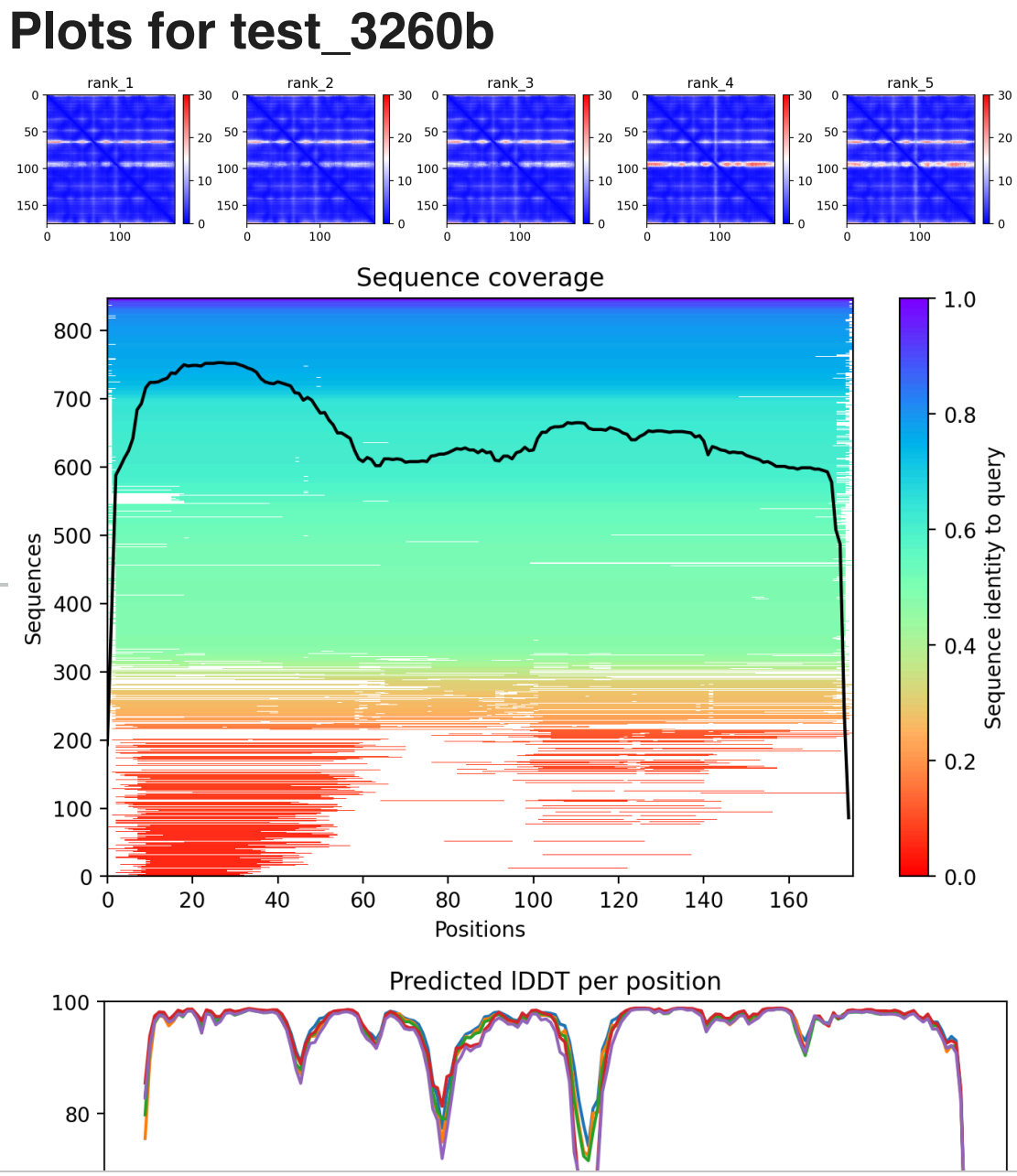
Show code



plDDT: Very low (<50) Low (60) OK (70) Confident (80) Very high (>90)

Plots

Show code



Package and download results

If you are having issues downloading the result archive, try disabling your adblocker and run this cell again. If that fails click on the little folder icon to the left, navigate to file: `jobname.result.zip`, right-click and select "Download" (see screenshot).

Show code

Instructions

For detailed instructions, tips and tricks, see recently published paper at [Nature Protocols]

Quick start

1. Paste your protein sequence(s) in the input field.
2. Press "Runtime" -> "Run all".
3. The pipeline consists of 5 steps. The currently running step is indicated by a circle with a stop sign next to it.

Result zip file contents

1. PDB formatted structures sorted by avg. pLDDT and complexes are sorted by pTMscore. (unrelaxed and relaxed if `use_amber` is enabled).
2. Plots of the model quality.
3. Plots of the MSA coverage.
4. Parameter log file.
5. A3M formatted input MSA.
6. A `predicted_aligned_error_v1.json` using AlphaFold-DB's format and a `scores.json` for each model which contains an array (list of lists) for PAE, a list with the average pLDDT and the pTMscore.
7. BibTeX file with citations for all used tools and databases.

At the end of the job a download modal box will pop up with a `jobname.result.zip` file. Additionally, if the `save_to_google_drive` option was selected, the `jobname.result.zip` will be uploaded to your Google Drive.

MSA generation for complexes

For the complex prediction we use unpaired and paired MSAs. Unpaired MSA is generated the same way as for the protein structures prediction by searching the UniRef100 and environmental sequences three iterations each.

The paired MSA is generated by searching the UniRef100 database and pairing the best hits sharing the same NCBI taxonomic identifier (=species or sub-species). We only pair sequences if all of the query sequences are present for the respective taxonomic identifier.

Using a custom MSA as input